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A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .

DOES MORE DATA (EXPERIENCE) LEAD TO BETTER PERFORMANCE?

“We don’t have better algorithms.
We just have more data!”

Peter Norvig—
Director of research, Google Inc



Further Reading #1

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Further Reading #1

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**Scaling to Very Very Large Corpora for
Natural Language Disambiguation**

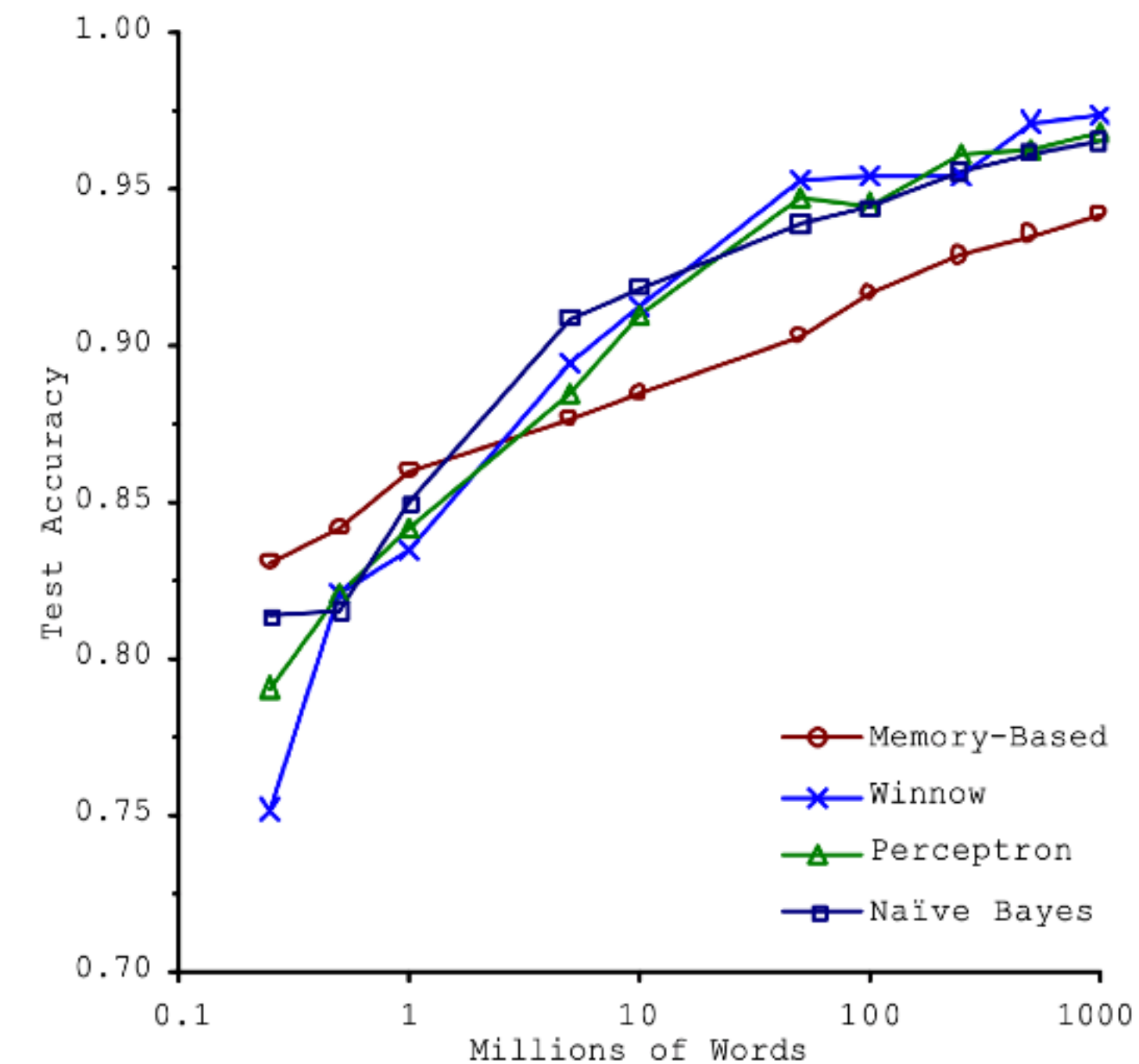
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Abstract

The amount of readily available on-line text has reached hundreds of billions of words and continues to grow. Yet for most core natural language tasks, algorithms continue to be optimized, tested and compared after training on corpora consisting of only one million words or less. In this paper, we evaluate the performance of different learning methods on a prototypical natural language disambiguation task, confusion set disambiguation when

potentially large cost of annotating data for those learning methods that rely on labeled text. The empirical NLP community has put substantial effort into evaluating performance of a large number of machine learning methods over fixed, and relatively small, data sets. Yet since we now have access to significantly more data, one has to wonder what conclusions that have been drawn on small data sets may carry over when these learning methods are trained using much larger corpora.

In this paper, we present a study of the effects of data size on machine learning for natural language disambiguation. In particular, we study the problem of selection among



“... We have shown that for a prototypical natural language classification task, the performance of learners can benefit significantly from much larger training sets. We have also shown that both active learning and unsupervised learning can be used to attain at least some of the advantage that comes with additional training data, while minimizing the cost of additional human annotation.”